Calculations based on Stoichiometry

1. How many moles of hydrogen is required to produce 10 moles of ammonia ?

   The balanced stoichiometric equation for the formation of ammonia is

   $N_2(g) + 3\ H_2\ (g) \rightarrow 2\ NH_3\ (g)$

   As per the stoichiometric equation,

   to produce 2 moles of ammonia, 3 moles of hydrogen are required

   ∴ to produce 10 moles of ammonia,

   $$\frac{3 \text{ moles of } H_2}{\cancel{2 \text{ moles of } NH_3}} \times \overset{5}{\cancel{10 \text{ moles of } NH_3}}$$

   = 15 moles of hydrogen are required.

2. Calculate the amount of water produced by the combustion of 32 g of methane.

   $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

   As per the stoichiometric equation,

   Combustion of 1 mole (16 g) CH_4 produces 2 moles (2 × 18 = 36 g) of water.

   CH_4

   $(12) + (4 \times 1) = 16\ g\ mol^{-1}$

   H_2O

   $(2 \times 1) + (1 \times 16) = 18\ g\ mol^{-1}$

   Combustion of 32 g CH_4 produces

   $$\frac{36 \text{ g } H_2O}{\cancel{16 \text{ g } CH_4}} \times \overset{2}{\cancel{32 \text{ g } CH_4}} = 72 \text{ g of water}$$

3. How much volume of carbon dioxide is produced when 50 g of calcium carbonate is heated completely under standard conditions?

   The balanced chemical equation is,

   $$CaCO_{3\,(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_{2\,(g)}$$

   As per the stoichiometric equation,

   1 mole (100g) $CaCO_3$ on heating produces 1 mole CO_2

   $CaCO_3$

   $(40) + (12) + (3 \times 16) = 100\ g\ mol^{-1}$

   At STP, 1 mole of CO_2 occupies a volume of 22.7 litres

   ∴ At STP, 50 g of $CaCO_3$ on heating produces,

   $$\frac{22.7 \text{ litres } of\ CO_2}{\underset{2}{\cancel{100 \text{g } CaCO_3}}} \times \cancel{50 \text{ g } CaCO_3}$$

   = 11.35 litres of CO_2

4. How much volume of chlorine is required to form 11.2 L of HCl at 273 K and 1 atm pressure ?

   The balanced equation for the formation of HCl is

   $H_2\ (g) + Cl_2\ (g) \rightarrow 2\ HCl\ (g)$

   As per the stoichiometric equation,

   under given conditions,

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